

Instant Coffee Tray Service SOP (1x Cup)

Prepare one cup of instant coffee per episode. Work in this order: move the saucer, place the cup, add coffee, pour water, stir, return the spoon, then move the cup and saucer together to the tray.

Arm assignments are fixed:

- **Left gripper:** moves the cup to center, holds the sachet for handoff, lifts the spoon at its middle for handoff, stabilizes the saucer during stirring, and returns the spoon to the plate.
- **Right gripper:** moves the saucer to center, takes, empties, and returns the sachet, handles the kettle, and takes the spoon at its handle end for stirring.
- **Final transfer:** both arms move the service together. The left gripper holds the saucer and the right gripper holds the cup handle.

Always grasp the cup by the middle of its handle. Never touch hot water or the kettle body, lid, or spout.

Setup

Start only after all input objects are separated on the left staging plate and both checklists pass.

Hardware checklist

1. Cameras are recording. The staging plate, center, kettle, and tray are visible.
2. The kettle spout and cup rim will remain visible during the pour.
3. Both arms are home with grippers open.
4. The kettle is closed and seated on its stand, handle toward the right arm.
5. The kettle contains at least 500 mL of water at 90–95 °C.
6. The table is clean, dry, and clear except for task materials.

Materials checklist

1. One clean, dry staging plate is on the left.
2. These objects are separated on the plate with visible gaps and no overlap:
 - one empty, dry 200 mL cup, upright with its handle facing right;
 - one dry saucer, face up;
 - one open coffee sachet, flat with its opening facing right; and
 - one dry teaspoon, bowl down with its handle facing right.
3. One empty, dry tray is in the right output zone.
4. The cup fits the saucer, and the cup and saucer fit fully on the tray.

Workspace layout

- **Left:** staging plate with the separated cup, saucer, sachet, and spoon
- **Center:** assembly and coffee-preparation area
- **Back center:** kettle and stand
- **Right:** serving tray

Vocabulary

- **Cup-handle grasp:** grasp the middle of the handle; never grasp the cup rim or body.
- **Centered cup:** an even ring of saucer is visible around the cup base.
- **Pour position:** the kettle spout is above the cup with a visible gap from the rim.
- **Correct fill:** liquid is 1–2 cm below the cup rim.
- **Dissolved coffee:** no granules or clumps are visible.
- **Unsafe spill:** hot water leaves the cup.

Steps

Step 1: Move the saucer, then the cup

Goal: the cup is centered on a stable saucer in the center.

1.1 Move the saucer

- With the **right gripper**, grasp the near saucer rim, lift it from the staging plate, and place it flat in the center.
- Confirm the saucer is stable. Reseat it once if it rocks.

1.2 Place the cup

- With the **left gripper**, grasp the middle of the cup handle.
- Lift the cup from the staging plate and lower it into the saucer well.

Check: the saucer is stable and the cup is centered. Regrasp the cup handle and reseat the cup once if needed. Do not slide either object.

Step 2: Hand off, empty, and return the sachet

Goal: the coffee is in the cup and the empty sachet is back on the staging plate.

2.1 Hold and hand off the sachet

- With the **left gripper**, hold the sachet at its lower end on the staging plate, opening up.
- With the **right gripper**, grasp the sachet body below the opening.
- Release the left gripper only after the right gripper has a secure grasp.

2.2 Empty and return the sachet

- With the right gripper, lift the sachet from the staging plate, keeping the opening up.
- Center the sachet opening inside the cup rim, invert it, and use two short vertical flicks.
- If granules remain, repeat the two flicks once.
- With the right gripper, return the empty sachet flat to its original staging-plate position, opening facing right.

Check: the sachet is empty and returned. Record any spill. Do not sweep granules into the cup.

Step 3: Pour hot water

Goal: fill the cup once to 1–2 cm below the rim and return the kettle.

3.1 Lift and position

- With the **right gripper**, grasp the kettle handle and lift it straight off the stand.
- Carry the kettle upright until the spout is in the pour position.

3.2 Pour

- Tilt and pour into the center of the cup. Stop at 1–2 cm below the rim.
- Return the kettle upright over the cup and hold for 2 seconds. Do not top up.

3.3 Return the kettle

- Carry the kettle upright to its stand, seat it fully, then release it.

Check: the kettle is stable and the fill is correct. Reseat a rocking kettle once. Do not correct an underfill or overfill.

Spill branch: if hot water leaves the cup, return the kettle upright and seat it if the path is safe. Clear both arms and end the episode. Do not move the cup or wipe hot water while recording. Use the station emergency procedure if the return path is unsafe.

Step 4: Stir and return the spoon

Goal: dissolve the coffee and return the spoon to the staging plate.

4.1 Lift and hand off the spoon

- With the **left gripper**, grasp the middle of the spoon and lift it from the staging plate.
- Present it at center. With the **right gripper**, grasp the end of the spoon handle.
- Release the left gripper only after the right gripper has a secure grasp.

4.2 Stir and inspect

- With the **left gripper**, hold the far saucer rim.
- With the **right gripper**, lower the spoon bowl into the coffee.

- Stir five full circles in one direction without touching the cup, then lift and drain the spoon over the cup for 2 seconds.
- If granules remain, repeat five circles once. Never perform a third cycle.

4.3 Hand back and return the spoon

- Release the saucer with the **left gripper**, then grasp the middle of the spoon.
- Release the right gripper only after the left gripper has a secure grasp.
- With the left gripper, return the spoon bowl down to its original staging-plate position, handle facing right.

Check: the coffee is dissolved, the cup remains centered, and the spoon is back on the plate.

Step 5: Move the cup and saucer together

Goal: the cup and saucer are stable on the tray.

5.1 Grasp and lift

- With the **left gripper**, grasp the near saucer rim.
- At the same time, with the **right gripper**, grasp the middle of the cup handle.
- Lift the cup and saucer together. Keep the saucer level and cup upright.

5.2 Transfer and release

- Carry both objects to the tray on separate, non-crossing arm paths.
- Lower the saucer into the marked tray position while the right gripper continues holding the cup.
- When the saucer is stable, release the saucer, then release the cup.

Check: the cup is centered, the saucer is stable, and both are fully inside the tray. Reseat the service once if it overhangs.

Step 6: End the episode

1. Confirm the cup and saucer are stable on the tray.
2. Confirm the kettle is seated and the sachet and spoon are back on the staging plate.
3. Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

Reset with recording off.

1. Let the drink and kettle reach a safe handling temperature.
2. Remove the service and discard the empty sachet.
3. Wash and dry all task objects, the table, and the kettle exterior.

4. Refill and heat the kettle, restage the four separated inputs, and run Setup.

SOP violations

These are actions that break the SOP and are reviewed in the side-by-side review tool.

How to record a violation

For each violation, record:

- the **start timestamp**;
- the **violation name** exactly as written; and
- the **SOP rule broken**.

Use the most specific label. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag the violation and retain the episode unless project policy excludes it. Record system and material failures separately.

Violations

Violation: Required sequence not followed

- **Visible cue:** the cup moves before the saucer; coffee, water, or stirring occurs out of order; or the service moves before the spoon is returned.
- **SOP rule broken:** Steps 1–6 (saucer, cup, sachet, kettle, spoon, final transfer, then end).
- **Coaching note:** follow the fixed sequence from saucer placement through final transfer.

Violation: Prescribed arm, gripper, or tool assignment not followed

- **Visible cue:** the left gripper moves the saucer, lifts or pours the sachet, handles the kettle, or stirs; the right gripper moves the cup; or the final transfer uses the wrong grips.
- **SOP rule broken:** Steps 1 to 5, left holds the sachet for handoff, right lifts, pours, and returns it, and all other actions use their written grippers.
- **Coaching note:** follow the written gripper assignment for each object and handoff.

Violation: Object handled or transported incorrectly

- **Visible cue:** an object is dragged, tilted when prohibited, moved on a crossing path, or released unstable.
- **SOP rule broken:** Steps 1–5 (lift before moving, follow the stated path and orientation, and release only when stable).
- **Coaching note:** use the stated grasp, path, orientation, and release condition.

Violation: Object dropped, collided, or released uncontrollably

- **Visible cue:** an object falls, strikes another object or arm, or is released outside its destination.
- **SOP rule broken:** Steps 1–5 (maintain control and release only at the assigned destination).
- **Coaching note:** confirm the grasp and landing area before moving.

Violation: Required check, retry, or recovery not followed

- **Visible cue:** a failed check is ignored, a retry limit is exceeded, or an unwritten recovery is used.
- **SOP rule broken:** Steps 1 to 5, perform each check and use only its allowed retry or recovery.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Cup not grasped by the handle

- **Visible cue:** a gripper closes on the cup rim or body, or away from the middle of its handle.
- **SOP rule broken:** Steps 1.2 and 5.1, always grasp the middle of the cup handle.
- **Coaching note:** always grasp the middle of the cup handle.

Violation: Saucer and cup moved in the wrong order

- **Visible cue:** the cup is moved before the saucer is stable in the center, or both are moved together from the staging plate.
- **SOP rule broken:** Steps 1.1–1.2 (place and stabilize the saucer before moving the cup).
- **Coaching note:** place the saucer with the right gripper, then the cup with the left.

Violation: Cup or saucer placed incorrectly

- **Visible cue:** the saucer rocks, the cup is off-center, or either object is slid instead of lifted for correction.
- **SOP rule broken:** Steps 1.1–1.2 (place the saucer flat and the cup centered; lift to correct).
- **Coaching note:** place the saucer flat and lift the cup by its handle to recenter it.

Violation: Sachet handled incorrectly

- **Visible cue:** the opening points down before reaching the cup; granules empty outside the cup; coffee remains inside; or the sachet is not returned flat.
- **SOP rule broken:** Step 2.2 (use the right gripper to empty fully into the cup and return the sachet to its original plate position).
- **Coaching note:** keep the opening up during transit, empty over the cup, then return the sachet.

Violation: Spill, debris, or waste handled incorrectly

- **Visible cue:** material lands outside the cup; granules are swept into the cup; or hot water is wiped or the hot service is moved after a spill.
- **SOP rule broken:** Step 2.2, Step 3 spill branch, and Step 4.2 (keep material in the cup and follow the prescribed spill response).
- **Coaching note:** do not return spills; follow the hot-water spill branch.

Violation: Pour safety condition not followed

- **Visible cue:** the kettle tilts before reaching the cup, the spout touches the cup, or the kettle is released before seating.
- **SOP rule broken:** Steps 3.1–3.3 (carry upright, keep a spout gap, and seat the kettle before release).
- **Coaching note:** carry upright, keep a gap, and seat before release.

Violation: Fill level out of range or cup topped up

- **Visible cue:** the liquid is outside the 1–2 cm range or a second pour is made.
- **SOP rule broken:** Step 3.2 (make one pour and stop 1–2 cm below the rim).
- **Coaching note:** stop in range on the first pour.

Violation: Handoff not completed correctly

- **Visible cue:** the left gripper does not hold the sachet before the right grasps it; the right lifts the sachet before the left releases; the spoon grips are incorrect; or a sending gripper releases before the receiving grasp is secure.
- **SOP rule broken:** Steps 2.1, 4.1, and 4.3 (complete each handoff with both grippers holding before the sending gripper releases).
- **Coaching note:** establish the receiving grasp before releasing the spoon.

Violation: Stirring performed incorrectly

- **Visible cue:** the left gripper stirs; fewer than five circles are made; direction reverses; the spoon touches the cup; the left gripper does not hold the saucer; or a third cycle is performed.
- **SOP rule broken:** Step 4.2 (left stabilizes the saucer; right stirs five circles in one direction, with one extension only when needed).
- **Coaching note:** hold with the left, stir with the right, and extend only once if needed.

Violation: Spoon not returned correctly

- **Visible cue:** the spoon remains in the cup or center, or is not bowl down in its original staging-plate position with its handle right.

- **SOP rule broken:** Step 4.3 (return the spoon bowl down to its original plate position, handle facing right).
- **Coaching note:** drain the spoon, then return it to its original plate position.

Violation: Final transfer not performed with both arms

- **Visible cue:** the cup and saucer move separately; one arm performs the transfer; the left gripper does not hold the saucer; or the right gripper does not hold the cup handle.
- **SOP rule broken:** Steps 5.1–5.2 (left holds the saucer and right holds the cup handle throughout the final transfer).
- **Coaching note:** grasp both objects and move them together with their assigned arms.

Violation: Final transfer or tray placement incorrect

- **Visible cue:** the arm paths cross; the service tilts; the cup or saucer is released early; or either object remains outside the tray.
- **SOP rule broken:** Steps 5.1–5.2 (use separate paths, keep both objects controlled, and release only after the saucer is stable on the tray).
- **Coaching note:** keep both objects controlled until the saucer is stable on the tray.

Violation: Workspace or tool state not restored

- **Visible cue:** the kettle is off its stand or the sachet or spoon is not in its plate position.
- **SOP rule broken:** Steps 2.2, 3.3, 4.3, and 6 (restore the sachet, kettle, and spoon before homing).
- **Coaching note:** restore all required final states before homing.

Violation: Incorrect episode ending

- **Visible cue:** recording ends before final checks, with arms away from home, with closed grippers, or without the 3-second hold.
- **SOP rule broken:** Step 6 (complete final checks, home both arms with open grippers, then end).
- **Coaching note:** complete final checks, hold, home both arms, then end.

Non-violation failures: log and discard an unusable episode for a recording, camera, robot, kettle, or fixture failure; defective material; or an unexpected safety hazard.

Annotation subtasks (from SOP)

1. Move the saucer to the center with the right gripper
2. Place the cup on the saucer with the left gripper
3. Hand the sachet from the left gripper to the right gripper
4. Empty and return the sachet with the right gripper

5. Lift, pour, and return the kettle with the right gripper
6. Hand the spoon from the left gripper to the right gripper
7. Stir and inspect the coffee
8. Hand the spoon back and return it to the staging plate
9. Grasp and move the cup and saucer together
10. Place the service on the tray
11. Return both arms home and end the episode